Weekly Seminar

Observability Analysis of Radar Inertial Odometry (RIO)

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Overview

1. Introduction

2. Main Results

3. Numerical Validation

4. Additional Results

5. Conclusion

Overall structure of the RIO

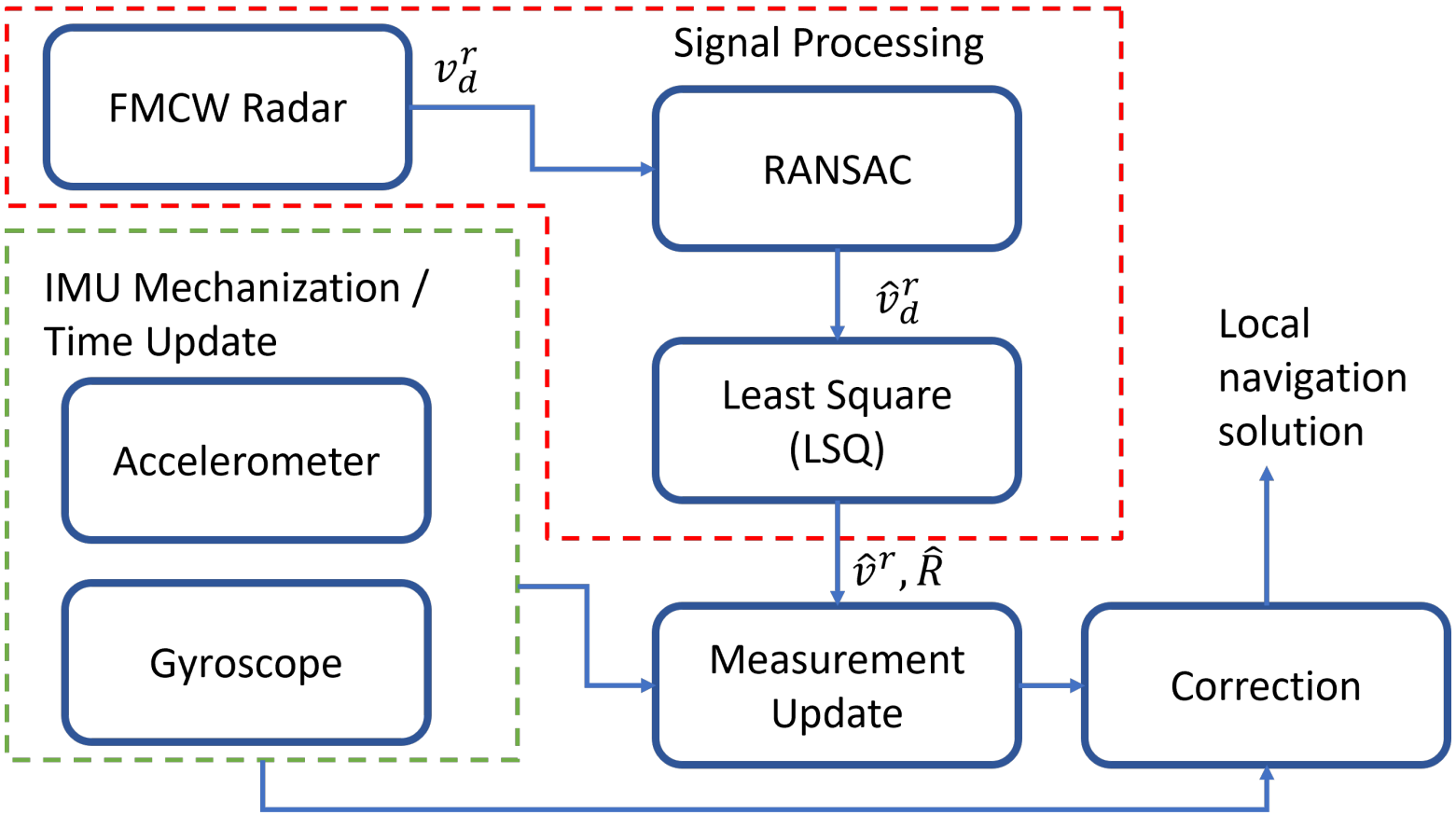


Figure 1: EKF-RIO block diagram.

IMU-Radar (RIO) Coordinate System

* We consider the following coordinate system: The navigation frame $\{N\}$, the IMU (body) frame $\{B\}$ and the radar frame $\{R\}$.
* The body frame rotate with the angular velocity ω^b.
* The homogeneous transform T includes the rotation matrix R and the translation vector p. For example: $T_b^n = \begin{bmatrix} R_b^n & p_b^n \\ 0_{1\times 3} & 1 \end{bmatrix}$.

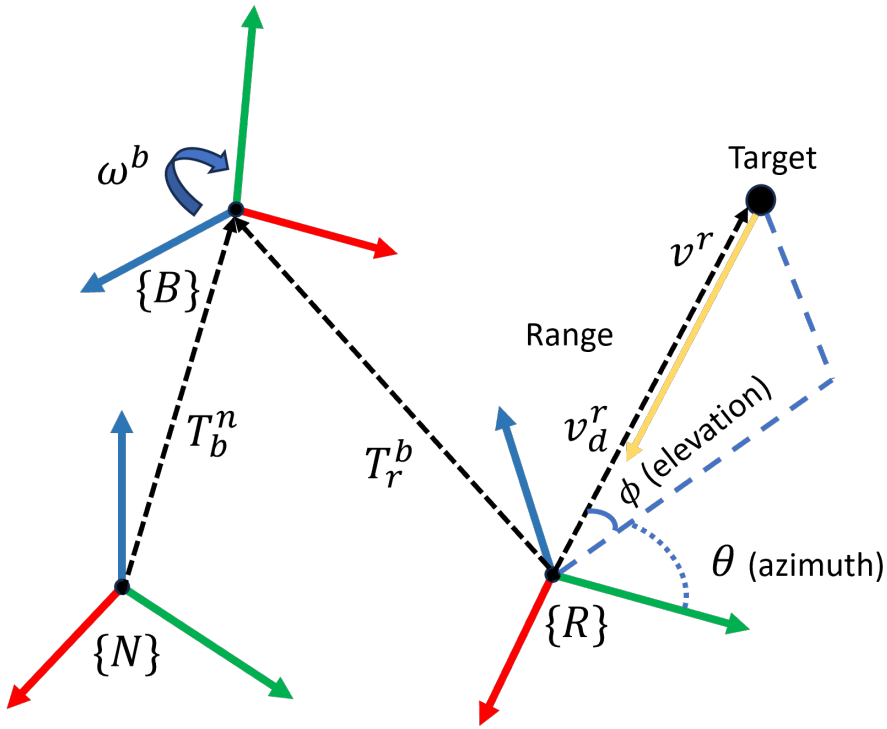


Figure 2: Sensor's coordinate definitions.

EKF-RIO

* The full state of EKF-RIO is defined as follows

$$x = \begin{bmatrix} {p_b^n}^T & {v_b^n}^T & {q_b^n}^T & b_a^T & b_g^T \end{bmatrix}^T \in \mathbb{R}^{16} \tag{1}$$

where p_b^n, q_b^n and b_a are respectively the IMU's position, Hamilton quaternion expressed in $\{N\}$ frame. The gyroscope and accelerometer biases are expressed in $\{B\}$ frame.

* The error state is define as follows

$$\delta x = \hat{x} - x = \begin{bmatrix} \delta p_b^n & \delta v_b^n & \delta\theta_b^n & \delta b_a & \delta b_g \end{bmatrix}^T \in \mathbb{R}^{15} \tag{2}$$

where $q = \delta q \otimes \hat{q}$ ($\otimes$ is the quaternion multiplication) and $\delta q \approx \begin{bmatrix} 1 \\ 0.5\,\delta\theta \end{bmatrix}$ ($\delta\theta$ is misalignment angle error). Thus, the linearized continuous-time error-state equation is[1]

$$\delta\dot{x} = F\delta x = \begin{bmatrix} 0_{3\times3} & I_3 & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & \left[\hat{C}_b^n \hat{f}^b\right]_\times & -\hat{C}_b^n & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & \hat{C}_b^n \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \end{bmatrix} \delta x \tag{3}$$

* The residual

$$r = R_n^b \hat{v}_b^n - \hat{v}^r = H\delta x \tag{4}$$

$$= \begin{bmatrix} 0_{3\times3} & H_v & H_q & 0_{3\times3} & H_{b_g} \end{bmatrix} \delta x \tag{5}$$

[1] C. Doer and G. F. Trommer, "An EKF Based Approach to Radar Inertial Odometry," 2020 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI), Karlsruhe, Germany, 2020.

Observability Analysis (1/2)

* The key reasons for observability analysis include[2]
  - Providing a deep insight about the system's geometric properties.
  - Determining the minimum measurement modalities to initialize the estimator.
  - Identifying degenerate motions.
  - Designing observability-based constrained filter to improve the consistency.
* Particularly, considering the discrete-time linear time-varying system

$$x_k = \Phi_{k-1} x_{k-1} \in \mathbb{R}^n \tag{6}$$

$$y_k = H_k x_k \in \mathbb{R}^m \tag{7}$$

  where $m < n$.
* The observability analysis indicates which component x_i of the state vector x_k (e.g., consider $x_k = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^\top \in \mathbb{R}^3$) can be estimated using the measurement y_k, and under what conditions.

[2] J. A. Hesch, D. G. Kottas, S. L. Bowman, and S. I. Roumeliotis. Consistency analysis and improvement of vision-aided inertial navigation. IEEE Transactions on Robotics, 30(1):158-176, Feb 2014.

Observability Analysis (2/2)

* To perform the analysis at time step k, we need to calculate the Gramian observability matrix

$$\mathcal{O}(x) = \begin{bmatrix} H_1\Phi_{1,1} \\ H_2\Phi_{2,1} \\ \vdots \\ H_k\Phi_{k,1} \end{bmatrix} \tag{8}$$

* The system is stated to be fully observable if $\text{rank}(\mathcal{O}(x)) = n$.
* However, since the matrix $\mathcal{O}(x)$ could have infinite row dimension, calculating the rank of $\mathcal{O}(x)$ could be burdensome.
* Another approach is to find the right nullspace matrix N such that[3].

$$\mathcal{O}(x)N = 0 \tag{9}$$

* The system is stated to be fully observable if the column rank of N is 0.

[3] Guoquan Huang, Anastasios I. Mourikis, and Stergios I. Roumeliotis. Analysis and improvement of the consistency of extended Kalman Filter-based SLAM. In Proc. of the IEEE International Conference on Robotics and Automation (ICRA), pages 473-479, Pasadena, CA, May 19-23 2008.

Lemma 1

With the initial condition $\Phi_{(1,1)} = I_{15}$, the exact closed-form discrete-time form $\Phi_{(k,k-1)}$ of $F(t_k)$ has the following structure[4]

$$\Phi_{(k,k-1)} = \left[\begin{array}{c|c|c|c|c} I_3 & \Delta t_k I_3 & \Phi^{(1,3)} & \Phi^{(1,4)} & \Phi^{(1,5)} \\ \hline 0_{3\times3} & I_3 & \int_{t_1}^{t_k} \left[\hat{C}_{b_\tau}^n \hat{f}^{b_\tau}\right]_\times \mathrm{d}\tau & -\int_{t_1}^{t_k} \hat{C}_{b_\tau}^n \mathrm{d}\tau & \Phi^{(2,5)} \\ \hline 0_{3\times3} & 0_{3\times3} & I_3 & 0_{3\times3} & \int_{t_1}^{t_k} \hat{C}_{b_\tau}^n \mathrm{d}\tau \\ \hline 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_3 & 0_{3\times3} \\ \hline 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_3 \end{array}\right] \tag{10}$$

where[5]

$$\Phi^{(1,3)} = \int_{t_1}^{t_k} \int_{t_1}^{s} \left[\hat{C}_{b_\tau}^n \hat{f}^{b_\tau}\right]_\times \mathrm{d}\tau\mathrm{d}s, \quad \Phi^{(1,4)} = -\int_{t_1}^{t_k} \int_{t_1}^{s} \hat{C}_{b_\tau}^n \mathrm{d}\tau\mathrm{d}s,$$

$$\Phi^{(1,5)} = \int_{t_1}^{t_k} \int_{t_1}^{\theta} \left[\hat{C}_{b_s}^n \hat{f}^{b_s}\right]_\times \int_{t_1}^{s} \hat{C}_{b_\tau}^n \mathrm{d}\tau\mathrm{d}s\mathrm{d}\theta, \quad \Phi^{(2,5)} = \int_{t_1}^{t_k} \left[\hat{C}_{b_s}^n \hat{f}^{b_s}\right]_\times \int_{t_1}^{s} \Phi^{(3,5)} \mathrm{d}\tau\mathrm{d}s$$

[4] Note that we are trying to find the analytic solution of the first-order differential equation $\dot{\Phi}_{(k,k-1)} = F(t_k)\Phi_{(k,k-1)}$.

[5] The cross product $a \times b = [a]_\times\, b, \forall a, b \in \mathbb{R}^3$ with $[a]_\times$ is a skew-symmetric matrix.

Proposition 1

The aided INS with Radar's body velocity measurement has **at least 4** *unobservable directions and the right nullspace subspace N_r of $\mathcal{O}$ is spanned by the following column basis function*

$$N_r = \begin{bmatrix} I_3 & 0_{3\times 1} \\ 0_{3\times 3} & \left[\hat{v}^n_{b_1}\right]_\times g^n \\ 0_{3\times 3} & g^n \\ 0_{3\times 3} & 0_{3\times 1} \\ 0_{3\times 3} & 0_{3\times 1} \end{bmatrix} \tag{11}$$

* Under a 6-DOF motion, the 3D global position x-y-z (first column of N_r) and the global rotation about the gravity direction (second column of N_r) are unobservable.
* The second unobservable direction (second column of N_r) is the linear combination of the horizontal error velocity parallel to $v^\perp = v^n_{b_1} \times g^n$ and the heading misalignment Ψ_D.

Hand-held carried board

* Sensor suite:
  1. Fast FMCW Radar TI IWR6843AOP.
  2. ADIS16448 IMU.
  3. Pseudo-groundtruth is generated by VINS with loop closure.
* C++ and ROS 2 (Foxy) - RIO module of DeRO[6].

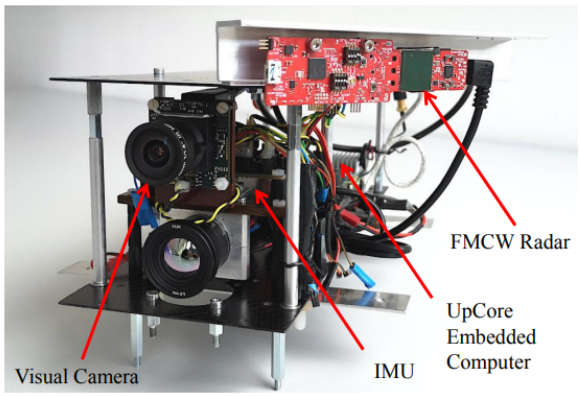
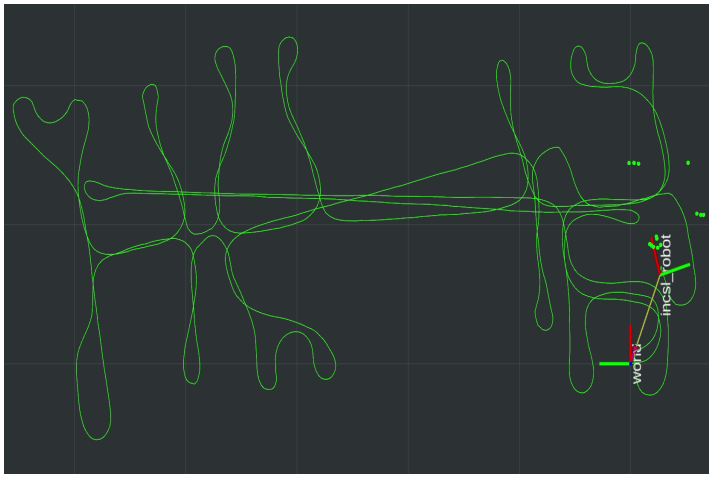


Figure 3: Drone's sensor platforms used for data collection[7].

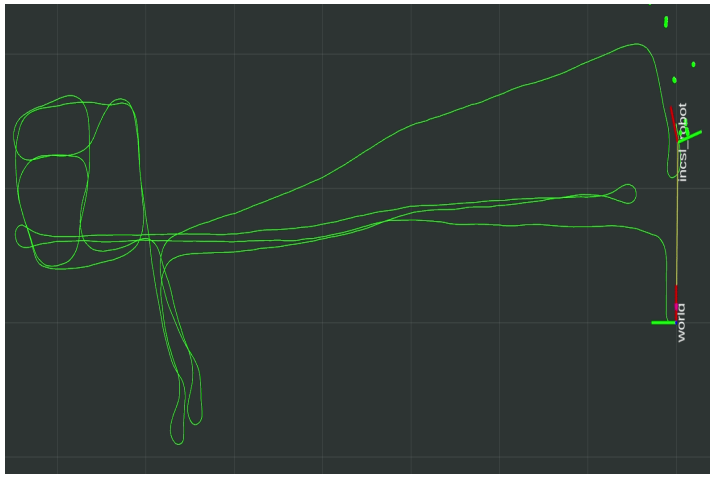
[6] Do, H. V., Kim, Y. H., Lee, J. H., Lee, M. H., & Song, J. W. (2024). DeRO: Dead Reckoning Based on Radar Odometry With Accelerometers Aided for Robot Localization. arXiv preprint arXiv:2403.05136. https://github.com/hoangvietdo/dero.

[7] C. Doer and G. F. Trommer, "Yaw aided Radar Inertial Odometry using Manhattan World Assumptions," 2021 28th Saint Petersburg International Conference on Integrated Navigation Systems (ICINS), Saint Petersburg, Russia, 2021, pp. 1-9, doi: 10.23919/ICINS43216.2021.9470842.

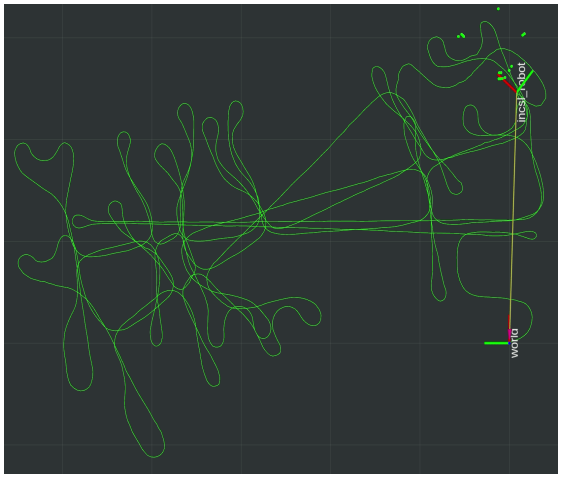
Datasets



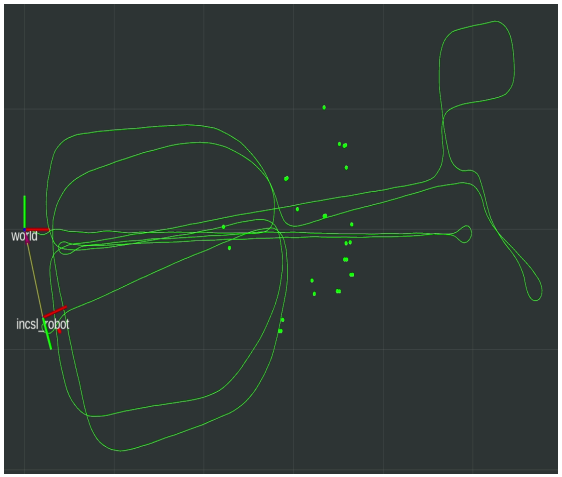
(a) Carried 1



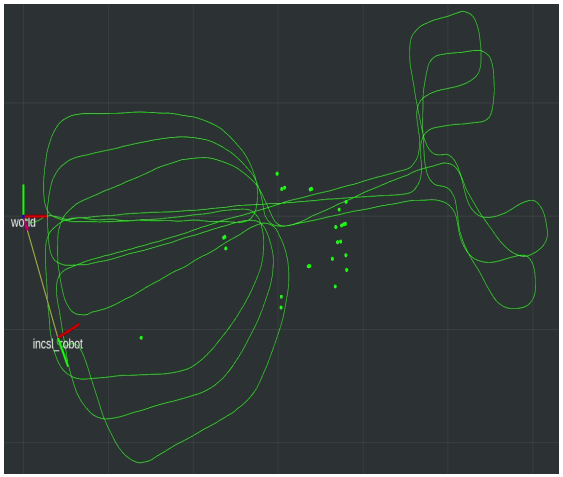
(b) Carried 5



(c) Carried 2



(d) Carried 3



(e) Carried 4

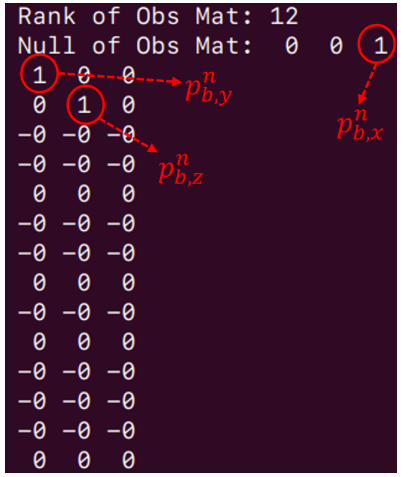
Figure 4: Five considered datasets.

Experiment Results (Partially reliable)

Table 1: Comparison of rank testing of the observability matrix $\mathcal{O}$ between theory and numerical method.

Dataset	EKF RIO	
	Theory	Numerical
Carried 1 (287 m, 273 s)	11	12
Carried 2 (451 m, 392 s)	11	12
Carried 3 (235 m, 171 s)	11	12
Carried 4 (311 m, 220 s)	11	12
Carried 5 (228 m, 172 s)	11	12

Figure 5: Example of numerical observability rank test of EKF-RIO for an arbitrary time instance k.



* The system is said to be *inconsistent* because we use the latest estimated point for linearization (standard EKF)[8].
* We can use observability-constrained based filter to solve this problem[9] [10].

[8] J. A. Hesch, D. G. Kottas, S. L. Bowman and S. I. Roumeliotis, "Consistency Analysis and Improvement of Vision-aided Inertial Navigation," in IEEE Transactions on Robotics, vol. 30, no. 1, pp. 158-176, Feb. 2014.

[9] G. P. Huang, A. I. Mourikis and S. I. Roumeliotis, "An observability-constrained sliding window filter for SLAM," 2011 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), San Francisco, CA, USA, 2011, pp. 65-72.

[10] Huang, G. P., Mourikis, A. I., & Roumeliotis, S. I. (2009). Observability-based Rules for Designing Consistent EKF SLAM Estimators. The International Journal of Robotics Research.

Degeneration Motion (1/4): Stationary Motions

Proposition 2

The aided INS with Radar's body velocity measurement with **stationary motion** *has* **7** *unobservable directions and the right nullspace subspace $N_{r,s}$ of $\mathcal{O}$ is spanned by the following column basis function*

$$N_{r,s} = \begin{bmatrix} I_3 & 0_{3\times1} & 0_{3\times1} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times1} & 0_{3\times1} & 0_{3\times1} & C_b^r \left[p_r^b\right]_\times \hat{C}_n^{b1} g^n \\ 0_{3\times3} & g^n & e_2 & -e_1 & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times1} & \hat{C}_n^{b1} g e_1 & \hat{C}_n^{b1} g e_2 & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times1} & 0_{3\times1} & 0_{3\times1} & \hat{C}_n^{b1} g^n \end{bmatrix} = \begin{bmatrix} N_1 & N_2 & N_3 & N_4 & N_5 \end{bmatrix} \tag{12}$$

where $e_i, i = 1, 2, 3$ are the basis function of $\mathbb{R}^3$ (e.g., $e_2 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^\top$).

* In this motion, $a = 0_{3\times1}, v = 0_{3\times1}, \omega = 0_{3\times1}$.
* The unobservable directions are global position (N_1), heading (N_2), x-y accelermoter biases (N_3 and N_4) and z gyroscope bias (N_5).
* Accelerometer biases in x and y direction are coupled with pitch and roll angle respectively (identical results with[11]).

[11] So Yeong Park. Fusion of the Integration and Parametric Approaches for PDR in Multiple Poses of Smartphone, Doctoral dissertation, Seoul National University, 2020.

Degeneration Motion (2/4): Stationary with Angular Velocity

Proposition 3

The aided INS with Radar's body velocity measurement with **only angular velocity excitation** *has* **5** *unobservable directions and the right nullspace subspace $N_{r,\omega}$ of $\mathcal{O}$ is spanned by the following column basis function*

$$N_{s,\omega} = \begin{bmatrix} I_3 & 0_{3\times 1} & 0_{3\times 1} \\ 0_{3\times 3} & 0_{3\times 1} & \frac{1}{\omega_z^2} C_b^r \left[p_r^b\right]_\times \hat{C}_n^b g^n \\ 0_{3\times 3} & g^n & 0_{3\times 1} \\ 0_{3\times 3} & 0_{3\times 1} & 0_{3\times 1} \\ 0_{3\times 3} & 0_{3\times 1} & \frac{1}{\omega_z^2} \hat{C}_n^b g^n \end{bmatrix} \tag{13}$$

* *Here, without loss of generality, we assume the sensor platform only rotates yaw angle.*
* *Without translation motion, one additional unobservable direction is gained (z axis gyroscope bias - last column of $N_{r,\omega}$) compared to a general motion.*
* *Rotating helps improving accelerometer biases estimation.*

Degeneration Motion (3/4): Constant Speed with no Angular Velocity

Proposition 4

The aided INS with Radar's body velocity measurement with **constant speed with no angular velocity** *has* **6** *unobservable directions and the right nullspace subspace $N_{r,v}$ of $\mathcal{O}$ is spanned by the following column basis function*

$$N_{r,v} = \begin{bmatrix} I_3 & 0_{3\times 1} & 0_{3\times 1} & 0_{3\times 1} \\ 0_{3\times 3} & 0_{3\times 1} & \left[v_{b_1}^n\right]_\times e_2 & -\left[v_{b_1}^n\right]_\times e_1 \\ 0_{3\times 3} & g^n & e_2 & -e_1 \\ 0_{3\times 3} & 0_{3\times 1} & \hat{C}_n^{b_1} g e_1 & \hat{C}_n^{b_1} g e_2 \\ 0_{3\times 3} & 0_{3\times 1} & 0_{3\times 1} & 0_{3\times 1} \end{bmatrix} \tag{14}$$

* *Without rotation motion, two additional unobservable directions are gained (x and y axis accelerometer biases) compared to a general motion.*
* *Velocity motions help improving gyro biases estimation in the z axis.*

Degeneration Motion (4/4): Other Possible Motions

Table 2: Number of unobservable modes for different motions and measurements of monocular Visual Inertial Navigation System (mono-VINS)[12].

Motion type	Point	Line	Plane
General	4	5	7
Only Translation	6 (Position + Attitude)	*	*
Only Rotation	5 (Feature Scale)	*	*
Constant Local Acceleration	5 (System Scale)	*	*
Moving Toward Points	5 (Feature Scale)	*	*
Moving Toward Line	*	6	*
Moving Along Line	*	6	*
Moving Perpendicular to Line	*	5	*

[12] Y. Yang and G. Huang, "Observability Analysis of Aided INS With Heterogeneous Features of Points, Lines, and Planes," in IEEE Transactions on Robotics, vol. 35, no. 6, pp. 1399-1418, Dec. 2019.

Scan Matching Measurement

Proposition 5

The aided INS with scan matching measurement (from Radar's pointcloud) has **at least** *4 unobservable directions and the right nullspace subspace N_s of $\mathcal{O}$ is spanned by the following column basis function*

$$N_s = \begin{bmatrix} I_3 & \left[\hat{p}^n_{b_1} + \hat{v}^n_{b_1}\Delta t\right]_\times g^n \\ 0_{3\times3} & \left[\hat{v}^n_{b_1}\right]_\times g^n \\ 0_{3\times3} & g^n \\ 0_{3\times3} & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times1} \\ I_3 & \left[\hat{p}^n_{b_{k-M}}\right]_\times g^n \\ 0_{3\times3} & g^n \end{bmatrix}, \quad \delta x = \begin{bmatrix} \delta p^n_{b,k} \\ \delta v^n_{b,k} \\ \delta \Psi^n_{b,k} \\ \delta b_{g,k} \\ \delta b_{g,k} \\ \delta p^n_{b,k-M} \\ \delta \Psi^n_{b,k-M} \end{bmatrix} \in \mathbb{R}^{21} \tag{15}$$

* Under a 6-DOF motion, the 3D global position x-y-z (first column of N_s) and the global rotation about the gravity direction (second column of N_s) are unobservable.
* Physical interpretation: The global position of the RIO at time k and $k-M$ (or feature points) move simultaneously (together at the same time in space) with the same amount.
* This result could be immediately applied (identical) to IMU-LiDAR[13] fusion system.

[13] J. Lv, X. Zuo, K. Hu, J. Xu, G. Huang and Y. Liu, "Observability-Aware Intrinsic and Extrinsic Calibration of LiDAR-IMU Systems," in IEEE Transactions on Robotics, vol. 38, no. 6, pp. 3734-3753, Dec. 2022.

Conclusion

* For the first time, we perform observability analysis for a FMCW Radar based inertial navigation system.
* The table below summarizes our results

Table 3: Number of unobservable modes for different motions and measurements of EKF-RIO.

Motion type	Body Velocity	Scan Matching
General	4	4
Stationary	7	*
Only Translation	6	*
Only Rotation	5	*

Thank you for listening

Q&A